Mahmoud Alyosify

Edge & Efficient AI Engineer | Inference Optimisation, Latency & Energy-Aware Deployment
MSc Artificial Intelligence, Queen's University (Canada)

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PROFESSIONAL SUMMARY

Efficient-AI Engineer whose entire focus is the constraint most ML work ignores: **making models cheap enough to actually run**. Master's research at **Queen's University, Canada** (expected **Nov 2026**, **Digilians Presidential Scholarship**) delivers a **training-free, zero-overhead decoding method** that cuts generated tokens by **25–35%** across three model families while measuring **TTFT, inter-token latency and GPU energy through NVML** — efficiency treated as a measured result, not a claim. Practical deployment experience with **ONNX export and FAISS-backed inference**, **4-bit quantised fine-tuning (QLoRA)**, **self-hosted model serving**, and **deep-RL resource provisioning under cost–latency budgets**. Systems background in **C/C++ and real-time national toll infrastructure**, plus a **200-hour ML internship with ITIDA / Egypt Makes Electronics**; currently an **AI Engineer Intern at Injaz Tech**, running a RAG system on a local 3B model on a 4 GB GPU. Seeking edge, on-device and inference-optimisation engineering roles.

EDUCATION

MSc in Artificial Intelligence — *Sep 2025 – Nov 2026*
Queen's University, Kingston, Ontario, Canada — Expected Nov 2026 · GPA: 4.30 / 4.30

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- **Master's Research Project (CISC 898):** *Verbosity-Aware Decoding for Token-Efficient Language Model Inference* — Supervisor: Dr. Rahatara Ferdousi.
- Coursework: **Large Language Models, Deep Learning, Cloud Computing, Reinforcement Learning, End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics) — *Sep 2019 – Jul 2023*
Assiut University, Egypt — GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — real-time contactless vital-sign estimation from video (rPPG). **Grade A+**; **Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

Inference Optimisation	Decoding-Time Token Reduction, Custom LogitsProcessors, Entropy-Based Gating, KV-Cache-Free Methods, Quantised Fine-Tuning (QLoRA / 4-bit), PEFT / LoRA Adapters, Self-Hosted Model Serving
Profiling & Measurement	Time-to-First-Token (TTFT), Inter-Token Latency, Throughput Benchmarking, **GPU Energy Measurement via NVML**, Deterministic Benchmarking on Pinned Hardware, Cost–Latency Trade-off Analysis, Per-Node Token & Cost Accounting
Deployment Runtimes	ONNX, FAISS Vector Search, Ollama (Local LLM Serving), Docker, FastAPI, Response Caching, Streamlit, AWS GPU Instances, Linux, CI/CD
ML & Deep Learning	PyTorch, TensorFlow, Hugging Face Transformers, CNNs, Self-Supervised & Contrastive Learning, Reinforcement Learning (PPO, DQN), Scikit-learn, Optuna
Systems & Low-Level	C, C++, Real-Time Application Support, File-System Implementation (Mini-FAT), Embedded-Oriented ML Training (ITIDA / Egypt Makes Electronics, 200 hrs), Git/GitHub, Bash
Programming	Python, C, C++, C#, SQL, Go, MATLAB, Bash

SELECTED EFFICIENCY & DEPLOYMENT PROJECTS

Verbosity-Aware Decoding — Zero-Overhead Token Reduction at Inference Time — *2026 – Present*
MSc Research Project (CISC 898), Queen's University · PyTorch · NVML · AWS GPU

- Built a **training-free LogitsProcessor** that cuts generated tokens at **decoding time** — **no retraining, no architecture change, no extra KV-cache or attention cost** — fusing stop-token, n-gram-repetition and semantic-redundancy signals under an **entropy-driven gate**.
- Instrumented the full efficiency picture rather than token count alone: **TTFT**, **inter-token latency** and **GPU energy draw via NVML**, all captured under deterministic runs on **pinned AWS GPU instances**.
- Cut generated tokens by **25.5–34.9%** on MT-Bench across all three models while meeting every pre-registered non-inferiority criterion.
- Proved the savings are not bought with quality loss, using **paired non-inferiority testing** (BERTScore, ROUGE, exact-match/F1, length-controlled LLM judge) across **Mistral-7B, Llama-3.1-8B** and **Qwen2.5-3B** on MT-Bench, XSum and SQuAD-v2.

SimCLR-Vision-SSL — From Trained Encoder to ONNX Retrieval Service *2026*
PyTorch · SimCLR · SupCon · ONNX · FAISS

- Exported a self-supervised vision encoder to **ONNX** and served its embeddings through a **FAISS** index, turning a research model into a runnable, framework-independent retrieval component.
- Reached **84.30% top-1 accuracy** after a **42-experiment augmentation sweep** plus a semi-supervised **SupCon** extension.

RL-Cloud-Autoscaler — Learned Provisioning Under a Cost–Latency Budget *2026*
PPO · DQN · Custom Gymnasium Environment · Resource Optimisation

- Built a custom simulation environment and trained **PPO** and **DQN** agents to allocate compute under an explicit **cost–latency reward**, benchmarked against static and threshold-based policies.

Horus-OSINT — Quantised Fine-Tuning & Self-Hosted Serving *2026*
Meta-Llama-3-8B · Fine-Tuning · AWS · Self-Hosted Inference

- Fine-tuned **Meta-Llama-3-8B** on **159,826** records and deployed it on **AWS** for self-hosted interactive inference — keeping model weights and data in-environment rather than calling a hosted API.

Additional Projects

- **AudioShield** — compact **CNN** over mel-spectrograms delivering real-time audio classification through a lightweight served application.
- **Vitalism Solution** — real-time video signal processing (rPPG + wavelet filtering) for contactless vital-sign estimation from a standard camera.
- **Also:** Mini-FAT File System (C, low-level systems) · HORUS Sentinel (multi-agent, self-hosted LLM) · SCRAP · Automated Multimodal Agent.

PROFESSIONAL EXPERIENCE

AI Engineer Intern *Oct 2026 – Present*
Injaz Tech — Injaz Group (formerly Fingerprint Consultancy) Maadi, Cairo, Egypt

- Running a **LangGraph** RAG system on a local **Qwen2.5-3B** model via **Ollama** on a 4 GB laptop GPU; pinning the context window to 8K moved the model from a 39/61 CPU–GPU split to **100% GPU** execution.
- Profiled tokens and latency per graph node (the relevance grader drove **75%** of token cost and 80% of time), replaced LLM calls with exact deterministic checks where possible, and added an answer cache cutting repeat-question latency from **34.7 s to 1.6 s**.

Machine Learning Instructor *Aug 2026 – Sep 2026*
National Telecommunication Institute (NTI) Arish & Assiut, Egypt

- Rehired to teach two **90-hour** applied ML bootcamps (**50 trainees**) at the Arish (online) and Assiut branches — regression, SVM, KNN, Naive Bayes, PCA and neural networks with Python labs, final exams and capstone-project evaluation.

Machine Learning Instructor *Jul 2025 – Sep 2025*
National Telecommunication Institute (NTI) Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA and neural networks with practical labs.

AI in Cybersecurity Intern *Jun 2025 – Jul 2025*
e& Egypt — via NTI (On-the-Job Training) Cairo, Egypt

- Applied AI-driven security tooling across threat intelligence, digital forensics and enterprise security analysis.

Systems Engineer (Military Service) *Feb 2024 – Mar 2025*
National Company for Roads Building & Development Cairo, Egypt

- Supported **real-time backend systems** for national toll infrastructure serving **50%+ of Egyptian vehicles** across **70%+ of national roads**, where latency and uptime were operational requirements, not metrics on a dashboard.
- Debugged and optimised live transportation software and supported legacy applications to minimise downtime; co-developed an internal data-management application in **C#, .NET, Entity Framework, SQL Server**.

Online Instructor — AI, Algorithms & Data Science *2020 – Present*
Udemy & YouTube Remote

- Built Arabic-language technical curricula reaching **10,000+ learners** and **700K+ YouTube views**.

CERTIFICATIONS & AWARDS

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)

Machine Learning Specialization — DeepLearning.AI

Digilians Presidential Scholarship — Queen's University

Semi-Finalist — Microsoft Imagine Cup 2023

AWS Academy — ML Foundations, ML for NLP, Data Engineering

HCIP – AI (Huawei)

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — Assiut University, 2022–2023

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency · **French:** Basic.